CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (EC) Examination

December 2012

EC 704 Digital Signal Processing & Applications

Date: 22.12.2012, Saturday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

SECTION – I

Q - 1 (a) The analog signal x(t) is given below [04]

$x(t) = 2\cos 2000\pi t + 3\sin 6000\,\pi t + 8\cos 12000\,\pi t$

Calculate: (i) Nyquist sampling rate?

(ii) If the given x(t) is sampled at the rate $f_s = 5000$ Hz. What is discrete time signal obtained after sampling?

(b) Write expression of linear convolution in terms of input sequence x(n), impulse response h(n) and response y(n). Derive that expression by taking any arbitrary discrete time sequence x(n). [03]

Q - 2 (a) State whether the following systems are Linear, Time-invariant, Causal, Stable [06]

$T(x[n]) = x[n] + 3u[n+1]$

(b) The system function of a causal linear time-invariant system is [08]

$$H(z) = \frac{1 - z^{-1}}{1 + \frac{3}{4}z^{-1}},$$

The input to this system is

$$x[n] = (1/3)^n\, u[n] + u[-n-1].$$

(a) Find the impulse response of the system h[n].
(b) Find the output y[n].
(c) Is the system stable?

OR

Q - 2 (a) State the properties of the Region of Convergence (R.O.C.) with suitable examples. [06]

(b) Compute linear convolution of following sequences using analytical method and tabular method of convolution. [08]

$x(n) = n/3$ for $0 \leq n \leq 6$ $\quad$ $h(n) = 1$ for $-2 \leq n \leq 2$

0 elsewhere. $\quad$ 0 elsewhere.

Q - 3 (a) Determine the Z-transform and sketch the R.O.C for following sequences. [07]

(i) $x[n] = (-1/3)^n u[n] - (1/2)^n u[-n-1]$

(ii) $x[n] = na^n u[n]$

(b) Explain Reconstruction of Band limited signal from its samples. [03]

(c) Explain polyphase decomposition in detail. [04]

OR

Q - 3 (a) Give decomposition of a stable, causal system into the cascade of a minimum-phase and an all-pass system; consider the two stable, causal systems specified by the system functions. [05]

$$H_1(z) = \frac{(1+3z^{-1})}{(1+\frac{1}{2}z^{-1})} \qquad H_2(z) = \frac{(1+\frac{3}{2}e^{j\pi/4}z^{-1})(1+\frac{3}{2}e^{-j\pi/4}z^{-1})}{(1-\frac{1}{3}z^{-1})}$$

(b) Explain Ideal frequency selective filters. Further explain phase distortion and group delay for the same. [05]

(c) Explain classifications of signal in brief. [04]

SECTION – II

Q - 4 (a) Draw and explain a radix-2 Decimation in time FFT implementation signal flow graph (butterfly diagram) for 8 point FFT. [06]

(b) What do you mean by adaptive filtering? [01]

Q - 5 (a) Determine DFT of the following sequences [08]

(i) $x[n] = 1/4$, for $0 \leqslant n \leqslant 2$
$\quad 0$, Otherwise

(ii) $x[n] = 1/5$, for $-1 \leqslant n \leqslant 1$
$\quad 0$, Otherwise

(b) Explain filter design by impulse invariance method in detail. [06]

OR

Q - 5 (a) Obtain cascade and parallel realization of the system function given by [06]

$$H(z) = \frac{1+\frac{1}{4}z^{-1}}{(1+\frac{1}{2}z^{-1})(1+\frac{1}{2}z^{-1}+\frac{1}{4}z^{-2})}$$

(b) Explain continuous time signal processing of discrete time signal in brief. [04]

(c) Compare FIR and IIR filter in brief. [04]

Q - 6 (a) Design type I chebyshev filter Given: [08]

(i) Passband ripple = 1db

(ii) Stop band Cutoff freq. $\Omega_s = 2000\pi$

(iii) Cutoff freq. passband $\Omega_p = 1000\ \pi$

(iv) Attenuation of 40dB for $\Omega \geqslant \Omega_s$

(b) How circular convolution is different from linear convolution. Explain with suitable example. [06]

OR

Q - 6 (a) Find inverse DFT of X[k] = {1, 2, 3, 4} [06]

(b) Explain direct form realization of IIR systems using direct form-I and direct form-II structures using constant coefficient difference equation [06]

$$y(n) = \sum_{k=1}^{N} a_k y(n-k) + \sum_{k=1}^{M} b_k x(n-k)$$

Where a_k and b_k are constants and $M \leq N$

(c) State and explain time shifting property of z-transform [02]

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